Jiayao (Claris) Gu

Last updated on October 23, 2026

jia.yao.gu@mail.mcgill.ca • (+1) 438-993-9637
LinkedIn • Ph.D. candidate @ Mila and @ McGill • Homepage @ https://jiayaogu.github.io

Biography

I am a Ph.D. candidate at **Mila – Québec AI Institute** and **McGill University**, specializing in **Generative AI**, **Large Language Models** and their applications on **Science**. Representative contributions include **(1)** a first-author paper introducing a dual-transformer-encoder framework accepted at ISMB (16% acceptance rate); **(2)** a co-first-authored paper on model-based optimization with Yoshua Bengio, awarded the Survey Certificate at TMLR 2026; and **(3)** a co-authored ICML 2026 spotlight paper (top 2.2%).

My research excellence is recognized through the **DNA2RNA Scholarship** ($80,000), the **FRQ-NT Scholarship** ($26,666) for genome-based therapies, the **Tinker Research Grant** ($5,000), and **Cohere Labs' Catalyst Grants**. Previously, I received my B.Sc. in **Computer Science and Biology** from McGill University, graduating on the **Dean's Honour List** (top 10%). Beyond academia, I have conducted agentic ai research and served as an intern supervisor at **Tiptree Systems**. I also received the *Rising Star Award* at the McMedHacks Hackathon for a 3D medical image segmentation challenge.

Education

Mila - Québec AI Institute / McGill University — **Sep 2024 – Present**
Ph.D. Computer Science (GPA: **3.86/4.0**) — Montréal, QC, Canada
Supervisor: Prof. Yue Li (homepage)
Sponsored by the prestigious **DNA2RNA Scholar Award** ($80,000) for exceptional graduates.

McGill University — **Sep 2022 – Sep 2024**
MSc. Computer Science — Montréal, QC, Canada
Awarded **FRQNT Research Scholarship** ($26,666) recognizing top Québec graduate students.
Received **Graduate Excellence Award** ($33,000) for high-performing graduate students.

McGill University — **Sep 2018 – Apr 2022**
BSc. Computer Science and Biology — Montréal, QC, Canada
Recognized on the **Dean's Honour List** (Top 10% of the faculty).

Experience

Tiptree Systems — **May 2026 - Jan 2027**
Intern supervisor — Montréal, Québec, Canada

- Offer valuable support with my expertise in machine learning to build agentic system that verify claims of paper reviewers.

Mila - Québec AI Institute | **McGill University** — **Jan 2024 - Present**
Research Assistant — Montréal, Québec, Canada

- Leading research in biological molecule synthesis and function prediction [2].
- Leading scientific agent development.

McGill University — **Aug 2021 - Aug 2024**
Graduate Researcher — Montréal, Québec, Canada

- Developed and optimized an 3D CT image segmentation workflow using state-of-the-art CNN framework, improving prediction accuracy by 25%.
- Published findings in high-impact peer-reviewed journals [7] [8].

Hackathon Team Leader — Montréal, Québec, Canada

- McMedHacks (2021): Developed an RCNN-based solution for 3D CT image segmentation, earning the Rising Star Award.
- McGill CodeJam (2022): Created a machine-learning backend for ad impression prediction, winning Broadsign challenge prize.

Preprints

* denotes co-first author.

[1] **Jiayao Gu**, Yue Li. A generative-AI framework for target-Specific MicroRNAs towards RNAi-based drug design (2026) bioRxiv 2026.05.07.723585; doi:https://www.biorxiv.org/content/10.64898/2026.05.07.723585v1 (Submitted to PLOS Computational Biology)

Publications

* denotes co-first author.

[2] **Jiayao Gu**, Can (Sam) Chen, Yue Li. MiRformer: A Dual-Transformer-Encoder Framework for Predicting MicroRNA-MRNA Interactions from Paired Sequences (2026) bioRxiv 2025.11.21.689769; doi: https://doi.org/10.1101/2025.11.21.689769 (**Accepted to ISMB, acceptance rate=65/408=16%**)

[3] Zipeng Sun, Can Chen, Ye Yuan, Haolun Wu, **Jiayao Gu**, Christopher Pal, Xue Liu. Training Diffusion Language Models for Black-Box Optimization (2026) arXiv, https://openreview.net/forum?id=Z7wI0sor6i&referrer=%5Bthe%20profile%20of%20Xue%20Liu%5D%28%2Fprofile%3Fid%3D~Xue_Liu1%29 (Accepted to **ICML2026 Spotlight Top 2.2%**)

[4] Minsu Kim* , **Jiayao Gu*** , Ye Yuan, Taeyoung Yun, Yoshua Bengio, Can (Sam) Chen. Offline Model-Based Optimization: Comprehensive Review (2025). TMLR: https://openreview.net/pdf?id=QcSZWo1TLl (**Awarded Survey Certificate for papers that provide an exceptionally thorough survey of the topic**)

[5] **Jiayao Gu**, Yihong Li. The Role of Data Selection in Language Model Training (2025). ICML 2025 Workshop: https://arxiv.org/abs/2501.03826

[6] **Jiayao G.***, Kexin C.*, Peidong L., Yue, Y., Lynn A., Yang L., Tianyu S. ClawBot-Matching: Bidirectional, Explainable, and Learnable Collaboration Matching in Mixed Human-Agent Networks (2026). RLEval: Methods and Reinforcement Learning Environments for Evaluating AI Agents. https://openreview.net/forum?id=6NTH0H5Loo

[7] Jamie Magrill, Dan Moldoveanu, **Jiayao Gu**, Matieu Lajoie, Ian R. Watson. Mapping the Single Cell Spatial Immune Landscapes of the Melanoma Microenvironment (2023). Clinical & Experimental Metastasis: https://link.springer.com/article/10.1007/s10585-023-10252-4

[8] Magrill J*, **Gu Jiayao***, Allard-Puscas S., Ramsay L., Lajoie M., Karimi E., Maritan S.M., Dankner M., Rezanejad M, Zuo D.M., Wei Y.H., Petrecca K., Guiot M.C., Routy B., Siegel P.M, Quail D., Walsh L., Liviu C., Mihalcioiu D., Stagg J., and Watson IR. Spatially Mapping the Single-Cell Immune Landscapes of Melanoma Brain Metastases Using Imaging Mass Cytometry (2024). Journal of Clinical Oncology: https://ascopubs.org/doi/10.1200/JCO.2024.42.16_suppl.e21517

Awards and Scholarships

D2R Doctoral Scholar Award 2026
Amount: $80,000
Scope: National
Description: Doctoral Award for exceptional graduate researchers in Canada.

ISMB 2026 Conference Fellowship 2026
Amount: $750
Scope: Global
Description: Fellowship awarded to students to present their papers at ISMB conference.

Laura Bassi Scholarship 2026
Amount: $500
Scope: Global
Description: Award to support exceptional work undertaken by all graduates.

DNA2RNA (D2R) Scholar Award 2025
Amount: $27,000
Scope: Provincial
Description: **Highest recognition for D2R trainees**; awarded to outstanding and active researchers in genomic-based therapeutics.

FRQNT Research Scholarships 2023
Amount: $26,666
Scope: Provincial (Québec)
Description: Research Scholarships sponsored by Fonds de recherche du Québec - Nature et Technologies (FRQNT) by the Québec government, recognizing **excellent graduate students** in Québec.

Grad Excellence Award 2022
Amount: $33,000
Scope: Institutional (McGill University)
Description: Awarded by McGill University to top-performing graduate students.

McGill CodeJam Challenge Winner 2022
Amount: $600
Scope: Institutional (McGill University)
Description: **1st Place Winner** of the Broadsign machine learning challenge for ad impression prediction.

Rising Star Award in 2021 McMedHacks Medical Image Hackathon 2021
Amount: $100
Scope: Institutional (McGill University)
Description: McMedHacks 2021 Medical Image Hackathon for 3D CT image segmentation innovation

Dean's Honour List (Top 10%) 2019
Scope: Institutional (McGill University)
Description: Recognized as the top 10% in the faculty based on CGPA.

Computer Science Skills

Programming: Python, Java, C, C++, R, bash scripting, HTML
Machine Learning Frameworks: PyTorch, TensorFlow, JAX, scikit-learn
Cloud Computing: GCP, AWS
Tools: Skilled in using Git/GitHub, MATLAB